# HK IMO Training 2019-2020
## Problem Set 22
### For 11 April 2020



**Problem 85.** Suppose that positive numbers are written in the squares of a $10 \times 10$ table. Frogs sit in five squares and cover the numbers in these squares. Alex calculated the sum of all visible numbers and got $10$. Then each frog jumped to an adjacent square and Alex's sum changed to $10^2$. Then the frogs jumped again to adjacent squares, and the sum changed to $10^3$ and so on. Every new sum became $10$ times greater than the previous one.
What is the maximum sum Alex can obtain?

**Problem 86.** Altitudes $AH_a$, $BH_b$, $CH_c$ of an acute triangle $ABC$ meet at $H$. Let $M_a$, $M_b$, $M_c$ be the midpoints of $BC$, $CA$, $AB$, respectively. Points $A'$, $B'$, $C'$ on the segments $AH$, $BH$, $HH_c$, respectively, are such that $\angle M_aB'A' = \angle M_bC'B' = \angle M_cA'C' = 90°$.
Prove that the lines $AC'$, $BA'$, $CB'$ are concurrent.

**Problem 87.** A connected graph is given. Prove that its vertices can be coloured blue and green and some of its edges marked so that

(i) every two vertices are connected by a path of marked edges,
(ii) every marked edge connects two vertices of different colour, and
(iii) no two green vertices are connected by an edge of the original graph.

**Problem 88.** For an integer $n > 1$, let $g(n)$ denote the greatest prime factor of $n$. A *strange pair* is an unordered pair of distinct primes $p$ and $q$ such that $\{p, q\} = \{g(n), g(n + 1)\}$ for no integer $n > 1$.
Prove that there exist infinitely many strange pairs.